الطالب الأكاديمي
THE ACADEMIC STUDENT

Faculty of Business Studies

BUS101

Introduction to Math for Business

Final Examination / Version: B

2025-2026 / Spring

9 – June – 2026

Number of Exam Pages: (including this cover sheet)	**Time Allowed: 2 hours**

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Question 1: Choose the correct answer (10 marks)

1. Markup, margin, or gross profit is calculated as:

A) Cost − selling price
B) Selling price − cost
C) Selling price − operating expenses
D) Gross profit – overhead

2. If the selling price of an item is $227.24 and the markup is $68.17, what is the cost?

A) $159.07
B) $295.41
C) $68.17
D) $227.24

3. Convert a markup of 20% on selling price to its equivalent markup on cost

A) 28%
B) 30%
C) 22%
D) 25%

4. If an item is sold above cost but below the break-even point, the result is:

A) Gross profit
B) Reduced net profit
C) Operating loss
D) Absolute loss

5. Banker's interest calculations use which of the following?

A) 365 days
B) 360 days
C) 52 weeks
D) 30 day per month

6. A 6% annual interest rate compounded monthly for 2½ years results in how many compounding periods?

A) 12
B) 24
C) 30
D) 60

7. A nominal interest rate of 5% compounded semiannually means the interest rate per compounding period is:

A) 5%
B) 2.5%
C) 10%
D) 0.5%

8. In an ordinary annuity, payments are made:

A) At the end of each period
B) At the beginning of each period
C) Once per year only
D) At random intervals

9. A loan of $8,000 is taken for 1 year at 9% simple interest. What is the interest?

A. $720
B. $600
C. $900
D. $840

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10. A business borrows $25,000 for 9 months at 6% interest. How much is the interest?

A. $1,000 B. $1,125
C. $1,200 D. $1,500

Question 2: (Note: show the steps of calculation) (10 marks)

A. Solve for the missing numbers. Markup is based on cost. (6marks)

? %	C	$?
? %	M	$?
120%	S	$30.36

B. Find the percent of markdown and the amount of markdown. (4 marks)

Original Price	% Markdown	$ Markdown	Reduced Price
$34.10	? %	$?	$16.00

Question 3: (Note: show the steps of calculation) (10 marks)

A. Find the break-even point and the operating loss. (6 marks)

Cost	Operating Expense	Break-even Point	Reduced Price	Operating Loss	Absolute Loss
$43	$17	$?	$45	$?	none

B. A bakery bakes 150 loaves of bread for $2.20 each. 20% are wasted. If they want 25% markup on selling price, find the price per loaf. (4 marks)

Question 4: (Note: show the steps of calculation) (10 marks)

A. A dental office borrowed $30,000 at 6.5% interest on March 10. A payment of $10,000 was made on May 20. The loan term is 200 days. Find the balance owed after only the payment. (6 marks)

B. Use the Amortization Table to determine both the finance charge and the payment required to amortize a loan of $4600 at an annual interest rate of 16% with a term of 48 monthly payments. (4 marks)

Question 5: (Note: show the steps of calculation) (10 marks)

A. Find the amount of the following annuity due. Find the total interest earned. (6 marks)

Amount of each deposit	Deposited	Rate	Time (Years)	Amount of Annuity	Interest Earned
$ 7800	semiannually	4 %	9		

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B. Find the present value and the amount of interest earned. (4 mark)

Amount Needed	Time (Years)	Interest	Compounded	Present Value	Interest Earned
$ 11,600	9	4%	semiannually	$	$

End of the questions

LOAN AMORTIZATION TABLE (MONTHLY PAYMENT PER $1,000 TO PAY PRINCIPAL AND INTEREST ON INSTALLMENT LOAN)

Terms in months	7.50%	8%	8.50%	9%	10.00%	10.50%	11.00%	11.50%	12.00%
6	$170.34	$170.58	$170.83	$171.20	$171.56	$171.81	$172.05	$172.30	$172.55
12	86.76	86.99	87.22	87.46	87.92	88.15	88.38	88.62	88.85
18	58.92	59.15	59.37	59.60	60.06	60.29	60.52	60.75	60.98
24	45.00	45.23	45.46	45.69	46.14	46.38	46.61	46.84	47.07
30	36.66	36.89	37.12	37.35	37.81	38.04	38.28	38.51	38.75
36	31.11	31.34	31.57	31.80	32.27	32.50	32.74	32.98	33.21
42	27.15	27.38	27.62	27.85	28.32	28.55	28.79	29.03	29.28
48	24.18	24.42	24.65	24.77	25.36	25.60	25.85	26.09	26.33
54	21.88	22.12	22.36	22.59	23.07	23.32	23.56	23.81	24.06
60	20.04	20.28	20.52	20.76	21.25	21.49	21.74	21.99	22.24

Terms in months	12.50%	13.00%	13.50%	14.00%	14.50%	15.00%	15.50%	16.00%
6	$172.80	$173.04	$173.29	$173.54	$173.79	$174.03	$174.28	$174.53
12	89.08	89.32	89.55	89.79	90.02	90.26	90.49	90.73
18	61.21	61.45	61.68	61.92	62.15	62.38	62.62	62.86
24	47.31	47.54	47.78	48.01	48.25	48.49	48.72	48.96
30	38.98	39.22	39.46	39.70	39.94	40.18	40.42	40.66
36	33.45	33.69	33.94	34.18	34.42	34.67	34.91	35.16
42	29.52	29.76	30.01	30.25	30.50	30.75	31.00	31.25
48	26.58	26.83	27.08	27.33	27.58	27.83	28.08	28.34
54	24.31	24.56	24.81	25.06	25.32	25.58	25.84	26.10
60	22.50	22.75	23.01	23.27	23.53	23.79	24.05	24.32

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2%

PERIOD	TABLE 12-1 COMPOUND VALUE OF $1	TABLE 12-3 PRESENT VALUE OF $1	TABLE 13-1 AMOUNT OF ANNUITY OF $1	TABLE 13-2 PRESENT VALUE OF ANNUITY OF $1	TABLE 13-3 SINKING FUND VALUE OF $1
1	1.0200	0.9804	1.0000	0.9804	1.0000
2	1.0404	0.9612	2.0200	1.9416	0.4951
3	1.0612	0.9423	3.0604	2.8839	0.3268
4	1.0824	0.9238	4.1216	3.8077	0.2426
5	1.1041	0.9057	5.2040	4.7134	0.1922
6	1.1262	0.8880	6.3081	5.6014	0.1585
7	1.1487	0.8706	7.4343	6.4720	0.1345
8	1.1717	0.8535	8.5829	7.3255	0.1165
9	1.1951	0.8368	9.7546	8.1622	0.1025
10	1.2190	0.8203	10.9497	8.9826	0.0913
11	1.2434	0.8043	12.1687	9.7868	0.0822
12	1.2682	0.7885	13.4120	10.5753	0.0746
13	1.2936	0.7730	14.6803	11.3483	0.0681
14	1.3195	0.7579	15.9739	12.1062	0.0626
15	1.3459	0.7430	17.2934	12.8492	0.0578
16	1.3728	0.7284	18.6392	13.5777	0.0537
17	1.4002	0.7142	20.0120	14.2918	0.0500
18	1.4282	0.7002	21.4122	14.9920	0.0467
19	1.4568	0.6864	22.8405	15.6784	0.0438
20	1.4859	0.6730	24.2973	16.3514	0.0412
21	1.5157	0.6598	25.7832	17.0112	0.0388
22	1.5460	0.6468	27.2989	17.6580	0.0366
23	1.5769	0.6342	28.8449	18.2922	0.0347
24	1.6084	0.6217	30.4218	18.9139	0.0329
25	1.6406	0.6095	32.0302	19.5234	0.0312
26	1.6734	0.5976	33.6708	20.1210	0.0297
27	1.7069	0.5859	35.3442	20.7069	0.0283
28	1.7410	0.5744	37.0511	21.2812	0.0270
29	1.7758	0.5631	38.7921	21.8443	0.0258
30	1.8114	0.5521	40.5679	22.3964	0.0247
31	1.8476	0.5412	42.3793	22.9377	0.0236
32	1.8845	0.5306	44.2269	23.4683	0.0226
33	1.9222	0.5202	46.1114	23.9885	0.0217
34	1.9607	0.5100	48.0336	24.4985	0.0208
35	1.9999	0.5000	49.9943	24.9986	0.0200
36	2.0399	0.4902	51.9942	25.4888	0.0192
37	2.0807	0.4806	54.0340	25.9694	0.0185
38	2.1223	0.4712	56.1147	26.4406	0.0178
39	2.1647	0.4619	58.2370	26.9025	0.0172
40	2.2080	0.4529	60.4017	27.3554	0.0166
41	2.2522	0.4440	62.6098	27.7994	0.0160
42	2.2972	0.4353	64.8620	28.2347	0.0154
43	2.3432	0.4268	67.1592	28.6615	0.0149
44	2.3900	0.4184	69.5024	29.0799	0.0144
45	2.4378	0.4102	71.8924	29.4901	0.0139
46	2.4866	0.4022	74.3302	29.8923	0.0135
47	2.5363	0.3943	76.8168	30.2865	0.0130
48	2.5871	0.3865	79.3532	30.6731	0.0126

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